6 Two coils P and Q are placed close to one another, as shown in Fig. 6.1.

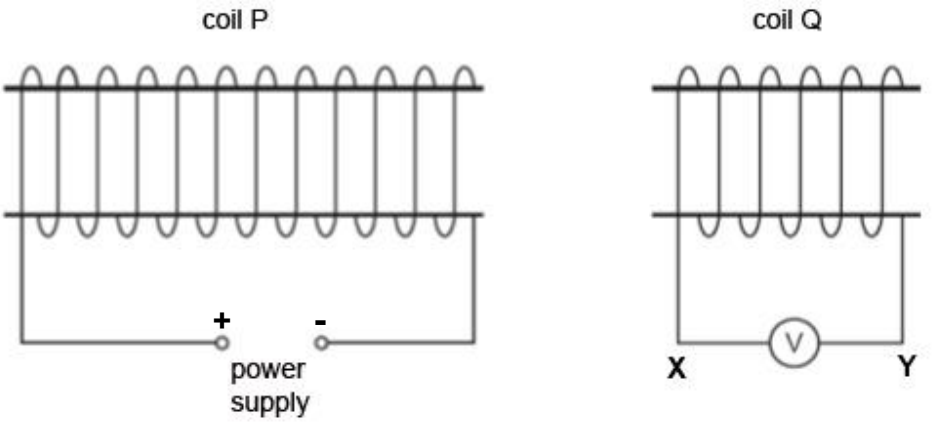

Fig. 6.1

(a) The current in coil P is constant. An iron rod is inserted into coil P.

During the time that the rod is moving,

(i) explain why, there is a reading on the voltmeter connected to coil Q.

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..... [2]

(ii) state and explain whether point X or point Y on coil Q is at a higher potential.

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..... [3]

(b) The current in coil P is now varied as shown in Fig. 6.2.

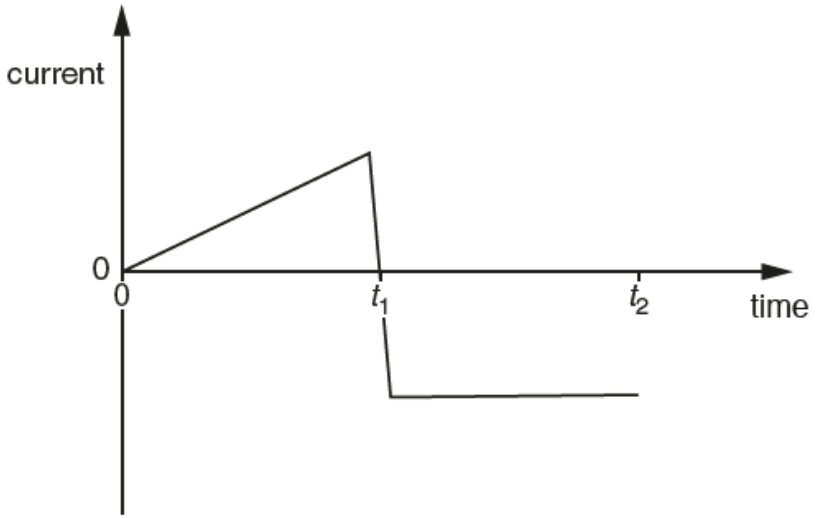

Fig. 6.2

On Fig. 6.3, show the variation with time of the reading of the voltmeter connected to coil Q for time $t = 0$ to time $t = t_2$.

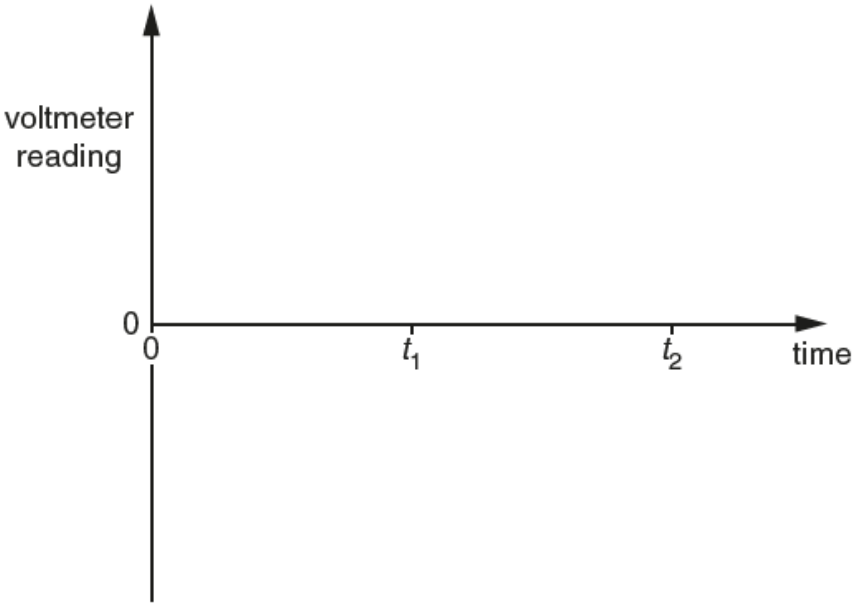

Fig. 6.3

[3]